

Glossary

Decision Tree, preprocessing, evaluation, oblique tree, and SHAP terms explained in simple language.

Learning Day 7 - Decision Trees for Customer Segmentation

Documentation generated from the finalized notebook and project results.

Glossary Overview

This glossary explains the main theory terms, metrics, model objects, and notebook variables used in the Decision Trees customer segmentation project.

Use this glossary while reviewing the notebook or preparing for interviews. It explains both machine learning concepts and actual project terminology.

A

Term	Meaning
Accuracy	The percentage of predictions that are correct. In this notebook, the baseline Gini tree test accuracy is 0.5105.
Actual segment	The true customer segment label in <code>y_test</code> .
Axis-parallel split	A standard Decision Tree split that uses one feature at a time, such as <code>Age <= 34</code> .

B

Term	Meaning
Baseline model	The first simple model used as a reference point. Here it is a <code>DecisionTreeClassifier</code> with <code>max_depth=3</code> .
BI-ready chart	A chart designed for business communication with clear title, subtitle, labels, readable colors, and business note.

C

Term	Meaning
Categorical feature	A feature made of categories or text values, such as <code>Gender</code> , <code>Profession</code> , or <code>Spending_Score</code> .
<code>ccp_alpha</code>	Cost-complexity pruning strength. Higher values prune the tree more strongly.
Class	A target category. In this notebook, classes are A, B, C, and D.
Classification	A supervised learning task where the model predicts a category.
Classification report	A table showing precision, recall, F1-score, and support for each class.
<code>ColumnTransformer</code>	A sklearn tool that applies different preprocessing steps to different column groups.
Confusion matrix	A table comparing actual classes against predicted classes.
Cost-complexity pruning	A tree simplification method that removes branches that do not add enough value.
Custom oblique tree	The custom expert model built with <code>LinearSVC</code> splitters inside a tree structure.

D

Term	Meaning
Data leakage	A mistake where information from the test set influences training. The notebook avoids this by fitting preprocessing only on <code>X_train</code> .
Decision node	A tree node that asks a question and splits rows into branches.
<code>DecisionTreeClassifier</code>	sklearn model class used to train standard Decision Trees.

E

Term	Meaning
Entropy	A split criterion that measures disorder in a node.

F

Term	Meaning
F1-score	The harmonic mean of precision and recall.
Feature	An input column used for prediction, such as Age or Profession_Artist.
Feature importance	A score showing how much each feature helped reduce impurity in the tree.

G

Term	Meaning
Gini impurity	A score measuring how mixed the classes are inside a node.

I

Term	Meaning
ID column	A unique identifier column. It is removed because it is not a behavioral feature.
Imputation	Filling missing values. Numeric values use median; categorical values use most frequent category.

L

Term	Meaning
Leaf node	A final node in a tree that gives a prediction.
LinearSVC	A linear support vector classifier used in the custom oblique tree to create linear-combination splits.

M

Term	Meaning
max_depth	Maximum number of levels the tree is allowed to grow.
min_samples_leaf	Minimum number of rows required in each final leaf node.
Multiclass classification	Classification with more than two classes. Here the target has four classes: A, B, C, D.

O

Term	Meaning
Oblique split	A split based on a linear combination of multiple features.
One-hot encoding	Converting a categorical column into binary columns, such as Profession_Artist.
Overfitting	When the model memorizes training data and performs poorly on test data.

P

Term	Meaning
Precision	Of all predictions made for a segment, how many were correct.
Processed feature names	The final feature names after imputation and one-hot encoding.

R

Term	Meaning
Recall	Of all actual rows in a segment, how many the model correctly found.
Root node	The first split in the tree. In this notebook, the root feature is Age.

S

Term	Meaning
Segment	The customer group label predicted by the model: A, B, C, or D.
SHAP	A method for explaining how features push predictions toward or away from a class.
SHAP beeswarm	A global plot showing which features influence predictions across many rows.
SHAP waterfall	A local plot explaining one prediction by adding feature effects from the baseline value.
SimpleImputer	sklearn preprocessing tool that fills missing values.
StandardScaler	A scaling method that transforms features to have comparable scale.
Stratified split	A train/test split that preserves class proportions in both sets.
Support	Number of true rows belonging to each class in the evaluation data.

T

Term	Meaning
Target variable	The column the model tries to predict. Here it is Segmentation.
Test data	Unseen data used to evaluate the trained model.
Train-test gap	Training accuracy minus test accuracy. A large positive gap may indicate overfitting.
Training data	Data used to fit the model.

U

Term	Meaning
Underfitting	When the model is too simple and fails to learn useful patterns.

V

Term	Meaning
Var_1	An anonymized categorical dataset column. It becomes one-hot columns such as Var_1_Cat_4.

W

Term	Meaning
Weighted Gini	Gini impurity after a split, weighted by the number of rows in each child node.

X

Term	Meaning
X	The feature matrix containing input columns.

Y

Term	Meaning
y	The target vector containing correct labels.